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Machine Learning Approach for Modeling Surface Tension Between H₂-CH₄ Gas Mixture and Brine: Implication for Underground Hydrogen Storage in Partially Depleted Gas Reservoirs

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Abstract

Hydrogen (H₂) has emerged as a pivotal component in the transition toward sustainable energy systems. However, large-scale deployment is constrained by the lack of sufficient storage infrastructure. Underground hydrogen storage (UHS) in depleted gas reservoirs is considered a promising solution due to its cost-effectiveness and scalability. Accurate estimation of interfacial properties, particularly surface tension (ST), is essential for assessing storage capacity and predicting fluid behavior in porous media. This study explores the use of machine learning (ML) models to predict ST between H₂/CH₄ gas mixtures and brine under reservoir-relevant conditions.

A dataset of experimentally measured ST values was used, incorporating input variables such as salinity, pressure, temperature, and density difference between gas and brine phases. Three artificial neural network (ANN) architectures—feedforward neural network (FFNN), legacy feedforward neural network (LFNN), and pattern recognition neural network (PRNN)—were first evaluated through an initial screening phase. Based on performance metrics, FFNN was selected as the most accurate model and was subsequently used for detailed sensitivity analysis. Two optimization algorithms, Levenberg–Marquardt (LM) and Bayesian Regularization (BR), were applied in the extended analysis. Additionally, the effect of the number of neurons and training epochs on model performance was assessed. Model accuracy was evaluated using R², mean squared error (MSE), and average absolute percentage error (AAPE), with training dynamics further analyzed using loss function plots.

The LM-optimized models consistently outperformed BR configurations, achieving the highest R² (0.986) during training and maintaining strong generalization in testing (R² = 0.982). BR models demonstrated robust performance, with all testing R² values exceeding 0.965. Among the tested cases, LM-20-100 (i.e., FFNN with LM optimization, 20 neurons, and 100 training epochs) offered the best balance of accuracy and computational efficiency. Loss function analysis revealed faster and smoother convergence in LM models, while BR models exhibited more gradual, multi-phase training behavior.

This study highlights the capability of ML, specifically FBNNs with LM and BR optimization, to accurately predict ST in complex gas–brine systems relevant to UHS. The results contribute to improving ST estimation under subsurface conditions, enhancing the screening and design of UHS operations in depleted gas reservoirs.

Keywords: Underground hydrogen storage, Hydrogen, Depleted gas reservoirs, Machine learning, Artificial intelligence

Introduction

Renewable energy sources play a central role in global strategies aimed at reducing carbon emissions by significantly lowering dependence on fossil fuels (Gielen et al., 2019). Among emerging energy carriers, hydrogen (H_2) stands out as a key component of the future energy landscape. In particular, green hydrogen—produced using renewable energy—offers a carbon-free alternative that supports widespread decarbonization efforts (Aggarwal and Briones, 2010; Ehteshami and Chan, 2014; Marouani et al., 2023). Its integration into the energy sector is especially valuable for lowering emissions in hard-to-decarbonize industries such as transportation (Barton and Gammon, 2010).

Despite its potential, hydrogen faces several technological and economic challenges throughout its value chain, including production (Younas et al., 2022), transportation (Li et al., 2022), storage (Felderhoff et al., 2007), and utilization (Aggarwal and Briones, 2010). One of the most pressing concerns is the lack of sufficient storage infrastructure for handling large hydrogen volumes, which poses a major obstacle to scaling up global hydrogen deployment (Akunets et al., 1994).

Hydrogen can be stored through physical or material-based methods (Nazir et al., 2020). Physical storage—such as compression and liquefaction—is commonly used but is energy-intensive and cost-prohibitive at large scales (Züttel, 2004; Cardella et al., 2017). Alternatively, underground hydrogen storage (UHS) offers a promising, cost-effective solution for large-scale, long-duration storage. UHS involves injecting hydrogen into suitable geological formations, including depleted oil and gas reservoirs, saline aquifers, and salt caverns (Muhammed et al., 2022; Perera, 2023).

By enabling long-term energy storage, UHS helps to balance supply and demand, enhance energy security, and integrate intermittent renewable energy sources into the grid (Amid et al., 2016; Heinemann et al., 2021). This capability is particularly critical for supporting decarbonization across sectors such as transportation and industry, where a stable and scalable hydrogen supply is needed (Gahleitner, 2013).

However, several technical challenges must be addressed to ensure the success of UHS. These include hydrogen leakage, the formation of hydrogen sulfide (H_2S), and corrosion due to microbial activity in subsurface environments (Dopffel et al., 2021; Rosman et al., 2023). A thorough understanding of the interactions among gas, water, rock, and microbial communities—as well as hydrogen's thermophysical behavior—is essential for safe and efficient storage operations (Tarkowski et al., 2021; Alanazi et al., 2024). Additionally, factors such as reservoir wettability, pressure, temperature, and pore geometry influence gas entrapment and transport mechanisms, all of which require further investigation (Bawazeer, 2022; Boon and Hajibeygi, 2022). In particular, the behavior of hydrogen at the pore scale in sandstone formations remains insufficiently explored (Jangda et al., 2022). For UHS to become commercially viable, these scientific and engineering gaps must be addressed to support large-scale investment and deployment (Watson et al., 2022).

The effectiveness of UHS is strongly influenced by the petrophysical and geo-mechanical properties of the storage reservoir, along with the interfacial characteristics of the gas–brine–rock system. Surface tension (ST) at the gas–liquid interface is a critical parameter for assessing hydrogen injectivity and trapping in depleted gas reservoirs, especially when mixed with cushion gases like methane (CH_4). Yet, experimental measurement of ST under reservoir conditions is both costly and time-consuming (Amooie et al., 2019; Nait Amar, 2021).

To overcome these limitations, researchers are increasingly turning to machine learning (ML) techniques. ML has shown significant promise in the oil and gas industry for modeling complex, heterogeneous systems and is now being applied to hydrogen storage challenges. These approaches offer accurate, scalable, and cost-effective alternatives to traditional experimental methods (Mehrijoo et al., 2020; Tariq et al., 2023).

Several studies have explored the use of ML models to estimate ST in hydrogen–brine systems, aiming to overcome the limitations of experimental methods. Azadivash (2024) employed ensemble regression models such as random forest (RF), Extra Trees (ET), gradient boosting (GB), Light Gradient Boosting Machine (LGBM), and CatBoost (CB) to predict ST in systems comprising H_2 and H_2/CO_2 gas mixtures with brine. Among these, the ET and GB models delivered the highest predictive accuracy, achieving R^2 values as high as 0.994, with RMSE and MAPE of 0.838 and 0.701, respectively.

Ng et al. (2022) developed ML models using genetic programming (GP), gradient boosting regression (GBR), and multilayer perceptron (MLP) to estimate ST based on input variables such as temperature, pressure, and molality. Among the models tested, the MLP optimized using the Levenberg–Marquardt (LM) and Adam algorithms achieved superior performance, yielding an R^2 of 0.999 and MAPE of 0.33. Similarly, Gbadamosi et al. (2024) utilized modern ML approaches—including Boosted Tree Algorithm (BTA), Elman Neural Network (ENN), Gaussian Process Regression (GPR), and Logistic Regression (LR)—to predict ST using temperature and density difference as inputs. The GPR model, when applied with the M2 input configuration, demonstrated exceptional accuracy, with a Nash–Sutcliffe Efficiency (NSE) of 1.0, RMSE of 0.0002, and MAPE of 0.0512.

Behnamnia et al. (2023) investigated ST prediction in systems containing hydrogen and various cushion gases (CH_4 , CO_2 , N_2), using an extensive dataset of 2868 points. Their models incorporated inputs such as temperature, pressure, salinity, and specific gravity. A wide array of algorithms was tested, including the Adaptive Neuro-Fuzzy Inference System (ANFIS), MLP models trained using Bayesian Regularization (BR), LM, and Scaled Conjugate Gradient (SCG), as well as hybrid methods based on the Grey Wolf Optimizer (GWO), such as GWO-LSBOOST, GWO-RBF, and GWO-LSSVM. Extreme Learning Machine (ELM) was also applied. The best-performing models in their study achieved an R^2 of 0.988, with RMSE and MAPE values of 1.38 and 1.70, respectively.

Previous ML-based studies on ST prediction often employed gas mixtures containing high fractions of cushion gases that are not typically representative of depleted gas reservoirs. For instance, some studies reported gas mixtures with CO_2 concentrations reaching up to 30 mol% (Ng et al., 2022), while others included N_2 fractions as high as 70 mol% (Behnamnia et al., 2023). Moreover, the dataset used by Behnamnia et al. (2023) included several scenarios in which H_2 was absent entirely. Such configurations may not accurately reflect the gas composition in depleted natural gas fields, where CH_4 is the dominant component.

In light of these limitations, the present study focuses specifically on predicting ST between H_2/CH_4 gas mixtures and brine under subsurface conditions, with the aim of supporting the assessment of UHS in repurposed depleted gas reservoirs. The dataset utilized in this work is derived from the experimental study conducted by Muhammed et al. (2023), which systematically measured the ST of CH_4/H_2 –brine systems under reservoir-relevant pressure, temperature, and salinity conditions.

To develop accurate predictive models, multiple types of ANNs were employed and evaluated. Based on the best-performing model, a detailed sensitivity analysis was conducted by tuning key hyperparameters, including the number of neurons in the hidden layer and the total number of training epochs. Two widely used optimization algorithms—Levenberg–Marquardt (LM) and Bayesian Regularization (BR)—were implemented to train the networks. The performance of each configuration was assessed using standard error metrics, and the results were thoroughly analyzed and compared. Additionally, loss function plots for all model combinations were generated and examined to provide deeper insight into the learning behavior and convergence trends of each network configuration.

Methodology

Data preprocessing

A total of 1,050 data points were obtained from the comprehensive experimental study conducted by [Muhammed et al. \(2023\)](#). The dataset includes key reservoir-relevant parameters such as salinity, pressure, temperature, and the densities of both the brine and the gas mixtures. ST values were also provided for each condition. For the purposes of model development, four primary input features were selected: salinity, pressure, temperature, and the density difference ($\Delta\rho$) between the gas and brine phases, as this contrast plays a critical role in interfacial behavior. The dataset was subjected to rigorous quality control to ensure accuracy, completeness, and consistency prior to its use in machine learning model training and evaluation.

To address the issue of varying data scales and to improve model convergence during training, the input data were normalized using a linear min–max scaling approach. This method transforms the original input values into a standardized range between -1 and 1 , thereby enhancing the stability and efficiency of the learning process. The normalization was performed using the following equation:

$$X_{normalized} = 2 \times \frac{X_{actual} - X_{min}}{X_{max} - X_{min}} - 1$$

This transformation ensures that all input variables contribute proportionately to the model's learning, while preventing dominance by features with inherently larger numerical ranges.

ML algorithms

In this study, several ANNs were employed, each consisting of multiple layers: an input layer, one hidden layer, and an output layer (see [Fig. 1](#)). The ANNs considered include the Feedforward Neural Network (FFNN), the Legacy Feedforward Neural Network (LFNN), and the Pattern Recognition Neural Network (PRNN).

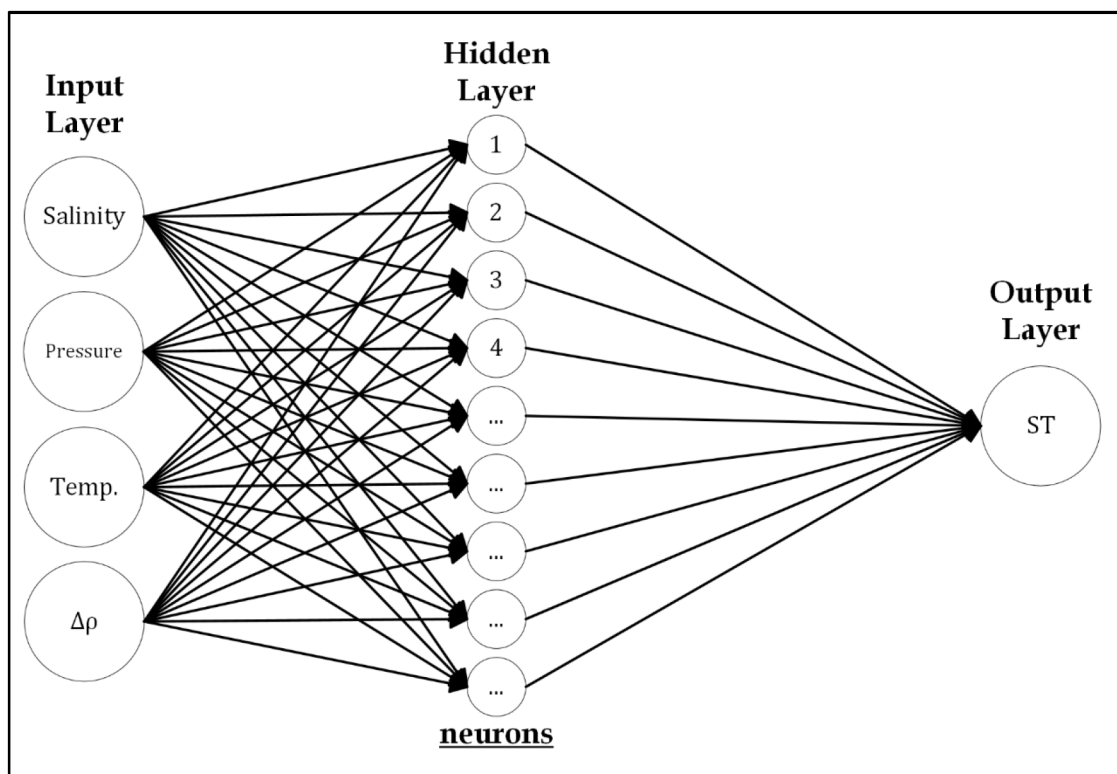


Figure 1—The general architecture of neural networks.

The FFNN is a type of neural network in which information flows strictly in one direction—from the input layer, through the hidden layer(s), to the output layer. Connections between neurons are unidirectional, with no feedback loops (Yalçın, 2021). An FFNN can be constructed with a flexible number of hidden layers and neurons depending on the complexity of the problem (Ma and Khorasani, 2003). This architecture has been widely applied in various domains, including pattern recognition, function approximation, dynamic system modeling, data mining, and time series forecasting (Xinghuo Yu et al., 2002).

The LFNN provides similar functionality to the FFNN but follows an older implementation. It has been successfully applied in earlier studies to model fluid and transport properties, making it a useful benchmark for comparative evaluation (Demuth et al., 2014).

The PRNN, originally developed for classification tasks, can be adapted for regression by appropriately configuring the output layer. Its inclusion in this study demonstrates the flexibility of neural network architectures for modeling continuously-valued outputs (Maier and Dandy, 2000).

Hyperparameters tuning

The initial three ANN architectures (i.e., FFNN, LFNN, and PRNN) were implemented using fixed configurations as part of a preliminary screening process to identify the most promising model for further investigation. Based on the results of this initial screening, the best-performing model was selected for detailed sensitivity analysis through systematic hyperparameter tuning.

To enhance the model's predictive accuracy, a structured optimization of key hyperparameters was conducted. Several parameters, such as the number of hidden layers (fixed at one) and the transfer function (softmax), were held constant throughout the analysis. The optimization process focused on three tunable components: the optimization technique, the number of neurons in the hidden layer, and the number of training epochs. Two widely used optimization methods were considered—Levenberg–Marquardt (trainlm) and Bayesian Regularization (trainbr). The number of epochs varied across 75, 100, and 125, while the hidden layer neuron count was tested at 10, 15, and 20. A summary of both the fixed and variable hyperparameter configurations used in the initial screening and sensitivity analysis is provided in Table 1.

Table 1—The hyperparameters optimization used in the ML model of this study for the initial screening and sensitivity analysis phases.

Parameter	Fixed	Variable	Values/Types
ANN types		x	fitnet (FFNN) [†] , newfit (LFNN), newpr (PRNN)
Number of training dataset	x		735 (70%)
Number of testing dataset	x		315 (30%)
Transfer function	x		softmax
Number of hidden layers	x		1
Number of epochs		x	75,100*,125
Learning function		x	trainlm (LM), trainbr (BR)*
Number of neurons per layer		x	10,15*,20

[†] Sensitivity analysis phase

* Initial screening phase

Model evaluation metrics

The predictive performance of the proposed ML models was assessed using three standard statistical metrics: the coefficient of determination (R^2), mean squared error (MSE), and average absolute percentage error (AAPE). These metrics quantify the degree of agreement between the experimentally measured ST values and the corresponding values predicted by the ML models.

The mathematical expressions for these evaluation metrics are given below:

$$R^2 = 1 - \frac{\sum_{i=1}^N (ST_{a,i} - ST_{p,i})^2}{\sum_{i=1}^N (ST_{a,i} - \bar{ST}_{a,i})^2}$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (ST_{a,i} - ST_{p,i})^2$$

$$AAPE = \frac{100}{N} \sum_{i=1}^N \left| \frac{ST_{a,i} - ST_{p,i}}{ST_{a,i}} \right|$$

Where $ST_{a,i}$ is the actual surface tension, $ST_{p,i}$ is the predicted surface tension, $\bar{ST}_{a,i}$ is the mean of actual surface tension, and N is the number of data points. These metrics were employed to evaluate the overall model accuracy and generalization performance across all model configurations.

Results and Discussion

Statistical Features

Fig. 2 presents violin plots for the five input and output variables used in the model, offering a comprehensive visualization of their statistical distributions (Molina et al., 2022). Violin plots are particularly effective for depicting the probability density of the data at different values, while simultaneously presenting key summary statistics.

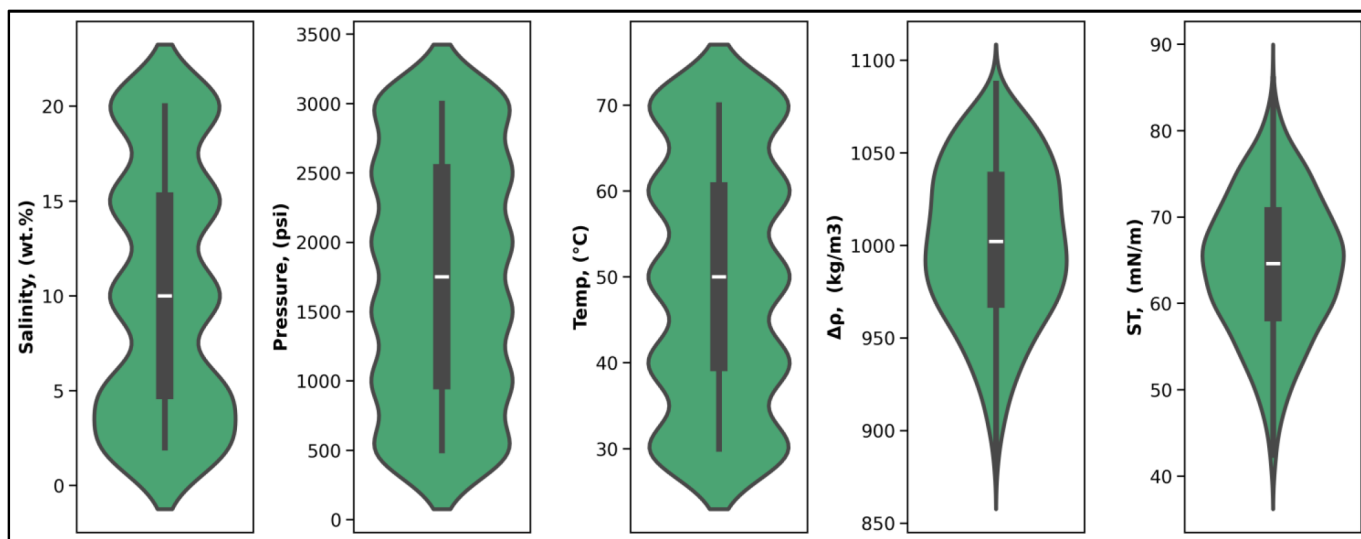


Figure 2—Violin plot of the basic statistics of model variables.

Each violin is symmetrical and displays a kernel density estimate on both sides, illustrating the distribution shape of each variable. The interquartile range (IQR) is represented by the central box within each violin, capturing the middle 50% of the data. A vertical black line extends from the lower to the upper whisker, indicating the overall data range, excluding potential outliers. The white horizontal tick mark inside each box denotes the median value of the corresponding variable.

This form of visualization enables a direct comparison of not only the central tendency and spread of each feature, but also the modality and skewness of their distributions. For instance, salinity and pressure show relatively wide distributions, while $\Delta\rho$ and ST appear more concentrated around their respective medians. Overall, the violin plots provide an intuitive, yet statistically informative summary of the model variables used in this study. They are especially useful for comparing multiple datasets or groups within a dataset, as they quickly reveal differences in central tendency, spread, and skewness. Detailed statistical parameters are listed in Table 2.

Table 2—Detailed statistical characteristics of model variables.

Parameter	Salinity, (wt.%)	Pressure, (psi)	Temp, (°C)	$\Delta\rho$, (kg/m ³)	ST, (mN/m)
Count	1050	1050	1050	1050	1050
Median	10	1750	50	1002	65
Minimum	2	500	30	878.6	40.2
Maximum	20	3000	70	1087.3	85.9
Average	10	1750	50	1001	64
SD	6.5	853.9	14.1	42.4	8.1
IQR	10	1500	20	64	11
Skewness	0.167	0.000	0.000	-0.232	-0.186

Fig. 3 presents the pairwise relationships and distribution profiles of the input and output variables used in the model. The diagonal elements of the plot represent kernel density estimates, capturing the underlying distribution of each feature. Notably, the most pronounced linear relationship is observed between $\Delta\rho$ and ST, as indicated by the concentrated diagonal pattern in their scatter plot. Additionally, a moderate association is observed between salinity and $\Delta\rho$, implying a potential interaction or partial dependence between these variables. These visual insights offer an initial indication of which features may have greater predictive relevance in modeling ST.

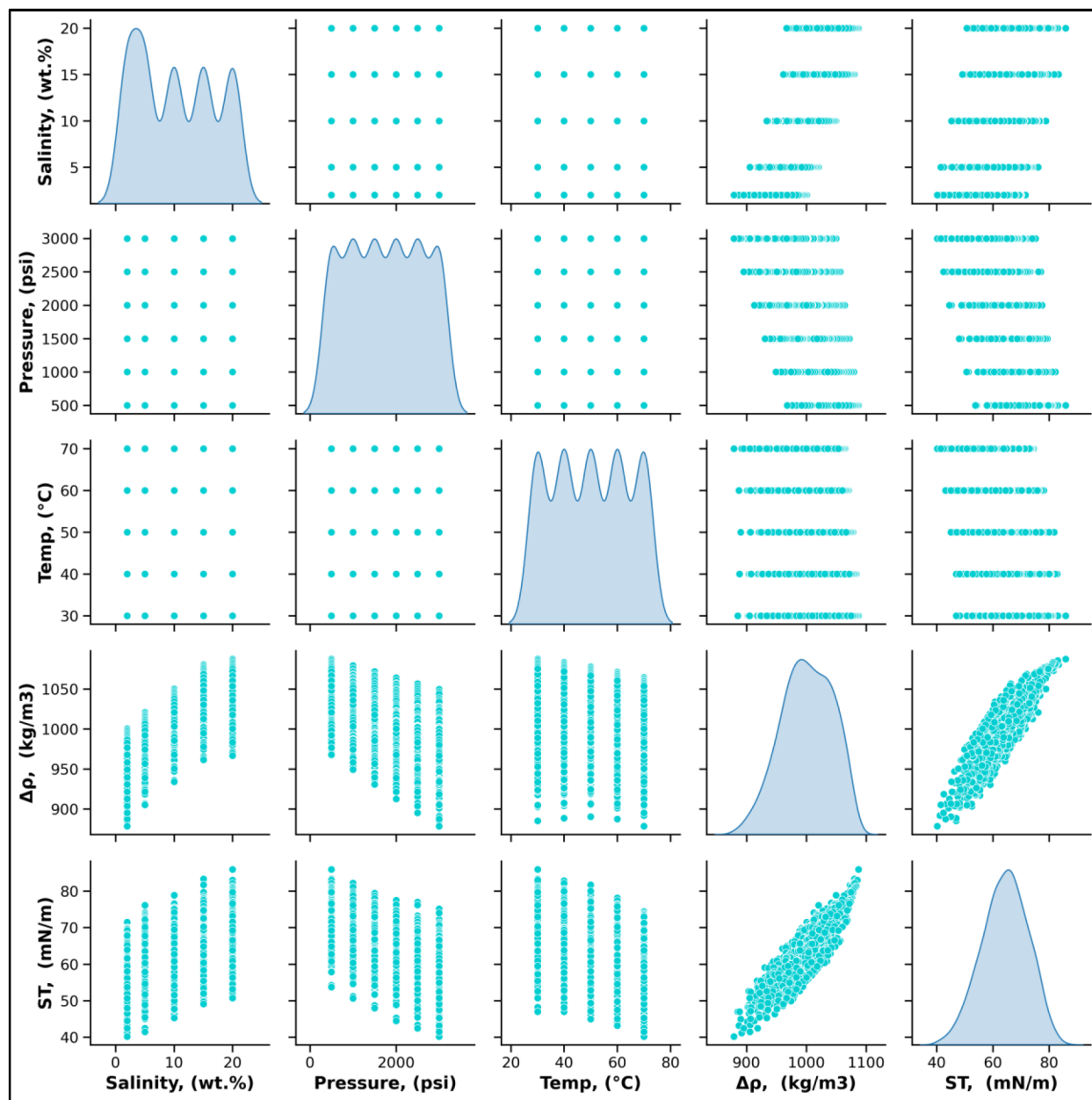


Figure 3—Pair plot of model variables that consists of cross-plots between all variables and probability density function plots of each variable (diagonal plots).

To further investigate the linear dependency of ST on individual input features, a tornado plot illustrating the correlation coefficients (CC) is presented in Fig. 4. The correlation coefficient ranges from -1 to $+1$, where values approaching ± 1 indicate strong linear relationships—positive or negative—while values near zero imply weak or no linear association. Each input variable exhibits a distinct level of linear correlation with ST. Among the features, $\Delta\rho$ shows the strongest positive correlation, with a CC of 0.90, suggesting it is the most influential predictor of ST in this dataset. Salinity and pressure also display moderate correlations, with CC values of 0.53 and -0.55 , respectively, indicating a direct and inverse linear relationship with ST. In contrast, temperature exhibits the weakest linear correlation, with a CC of -0.37 , reflecting limited predictive value in a linear context. These findings provide a preliminary understanding of variable importance and support the selection of $\Delta\rho$ as a key input feature in modeling performance.

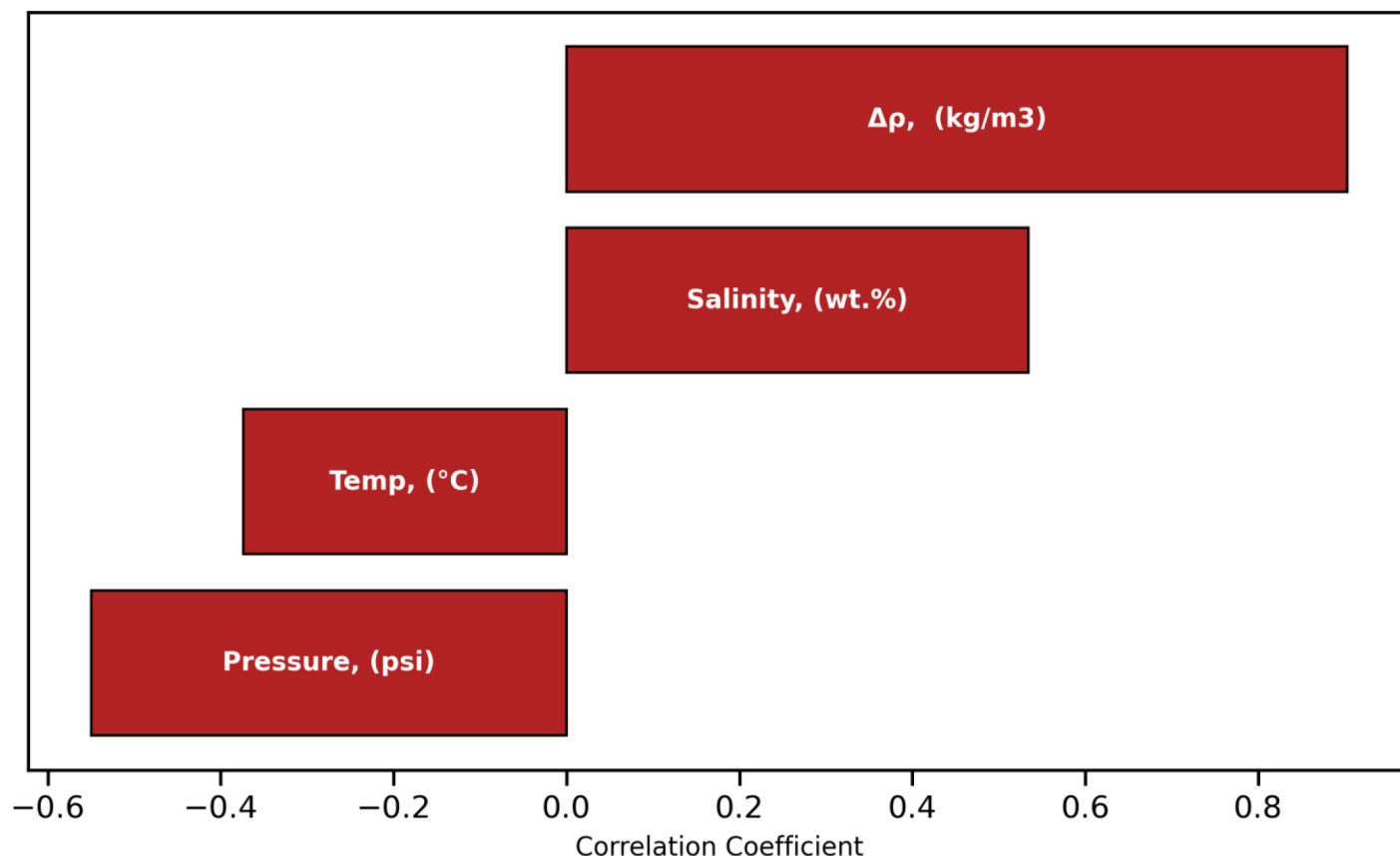


Figure 4—Heat map of correlation coefficient (R) between model inputs and ST.

Initial Screening

The dataset was initially divided into training and testing subsets using a 70:30 split, yielding 735 data points for training and 315 for testing the model's predictive performance. The considered ANN architectures were implemented and evaluated based on standard error metrics. Table 3 summarizes the performance of each model during both training and testing phases.

Table 3—The results of initial screening using multiple ANN models.

Model	R ²		MSE		AAPE	
	Training	Testing	Training	Testing	Training	Testing
FFNN	0.983	0.980	0.0021	0.0026	0.397	0.561
PRNN	0.979	0.974	0.0033	0.0039	0.484	0.606
LFNN	0.974	0.970	0.0027	0.0033	0.611	0.748
FFNN performance with respect to other models *						
PRNN	0.5%	0.6%	-22.2%	-21.2%	-17.9%	-7.3%
LFNN	1.0%	1.0%	-36.4%	-33.3%	-35.0%	-24.9%

* the percentage shows the relative change with respect to the other models (positive means higher and negative means lower)

Among the models tested, FFNN achieved the highest accuracy, with R² values of 0.983 for training and 0.980 for testing. Correspondingly, it reported the lowest errors, including a training MSE of 0.002 and a testing MSE of 0.003. However, in terms of AAPE, FFNN yielded values 35% and 18% lower than the training AAPE of LFNN and PRNN, respectively. For the testing phase, FFNN's AAPE remained relatively competitive, registering only 25% and 7% lower values than those of LFNN and PRNN, respectively.

PRNN ranked as the second-best performing model, with R^2 values closely trailing those of FFNN in both phases. LFNN, on the other hand, demonstrated the least favorable performance among the three models, making it less suitable for further analysis in this study.

Sensitivity Analysis

As detailed in the methodology section, various hyperparameters were systematically adjusted to assess their impact on the ML model's performance. The sensitivity analysis was conducted on the FFNN since it has shown the best accuracy in the initial screening.

Table 4 presents a summary of the model outcomes across different combinations of hyperparameters. Each model configuration is labeled according to its optimization method, number of neurons, and number of epochs. For example, the label LM-10-75 denotes a model trained using the Levenberg–Marquardt optimization with 10 neurons and 75 training epochs.

Table 4—Performance of optimized FFNN models in sensitivity analysis phase.

Model	R2		MSE		AAPE	
	Training	Testing	Training	Testing	Training	Testing
LM-10-75	0.979	0.973	1.3435	1.7812	0.0139	0.0161
LM-10-100	0.977	0.972	1.4669	1.8538	0.0147	0.0169
LM-10-125	0.978	0.973	1.4152	1.8146	0.0146	0.0167
LM-15-75	0.984	0.982	1.0277	1.2351	0.0124	0.0136
LM-15-100	0.974	0.969	1.7020	2.0456	0.0163	0.0181
LM-15-125	0.985	0.982	0.9777	1.2034	0.0123	0.0135
LM-20-75	0.973	0.970	1.7772	2.0192	0.0169	0.0180
LM-20-100	0.986	0.982	0.9166	1.1773	0.0117	0.0135
LM-20-125	0.986	0.982	0.8882	1.2119	0.0118	0.0140
BR-10-75	0.972	0.968	1.8518	2.1519	0.0170	0.0185
BR-10-100	0.973	0.968	1.7391	2.1255	0.0165	0.0187
BR-10-125	0.980	0.973	1.3038	1.7935	0.0138	0.0161
BR-15-75	0.972	0.969	1.8449	2.0726	0.0171	0.0182
BR-15-100	0.970	0.966	1.9602	2.2581	0.0176	0.0191
BR-15-125	0.971	0.968	1.8854	2.1137	0.0172	0.0181
BR-20-75	0.969	0.966	1.9911	2.2713	0.0176	0.0189
BR-20-100	0.971	0.968	1.8962	2.1315	0.0172	0.0183
BR-20-125	0.984	0.980	1.0500	1.3405	0.013	0.014

Among all configurations, the highest R^2 during training was achieved by LM-20-100 and LM-20-125, both reaching R^2 of 0.986. These findings suggest that extending training beyond 100 epochs offers no substantial gain in performance, indicating that 100 epochs is likely an optimal stopping point. Notably, all LM-10 models reported R^2 values below 0.98, while the best-performing LM-15 model achieved an R^2 of 0.985 when trained with 125 epochs.

In comparison, models trained with Bayesian Regularization (BR) displayed slightly lower R^2 values across most scenarios. The lowest training R^2 was observed for BR-20-75 (0.969), while the highest, BR-20-125, reached an R^2 of 0.984—still marginally below the top LM result. Overall, both LM and BR models demonstrated strong performance in the training phase.

However, the true test of any ML model lies in its ability to generalize to unseen data. High training accuracy may result from overfitting, or memorization, rather than true learning. Therefore, model performance on the testing dataset is a more critical indicator of robustness. Consistent with their training results, LM-20-100 and LM-20-125 also achieved the highest R^2 values during testing, each scoring 0.982. This consistency underscores their generalizability and practical applicability. Given that the performance gain from extending to 125 epochs is negligible, the 100-epoch model is preferred for its reduced computational cost.

On the other hand, BR-20-75 exhibited the lowest testing R^2 value of 0.966, which still reflects a good level of accuracy. In general, BR models showed slightly inferior performance compared to their LM counterparts in both training and testing phases, although all tested configurations demonstrated strong predictive capability suitable for ST modeling.

The remaining results in Table 4 present MSE and AAPE for all model configurations across both training and testing phases. Among the evaluated models, LM-20-125 achieved the lowest MSE during training, indicating the most precise fit to the training data. However, LM-20-100 outperformed all other models in the testing phase by yielding the lowest MSE, reinforcing its strong generalization capability.

Conversely, the highest MSE values in both training and testing were observed for BR-20-75, highlighting its relatively weaker performance compared to the other configurations. The AAPE results further corroborate the superior performance of LM-20-100, which exhibited the lowest AAPE values in both training (0.0117) and testing (0.0135), reflecting minimal average prediction error.

In contrast, the highest AAPE during training was reported for BR-15-100, while the most significant error during testing was again observed for BR-20-75. These findings consistently support the conclusion that the LM-20-100 model offers the best trade-off between accuracy and efficiency, making it the most reliable configuration for ST prediction under the tested scenarios.

Fig. 5 presents scatter plots comparing the actual ST values with the predicted values generated by the LM-based models for both training (Fig. 5-A) and testing (Fig. 5-B) phases. These plots offer a visual assessment of the model's predictive performance. The black dashed diagonal line in each plot represents the ideal case where predicted values perfectly match actual values (i.e., $R^2 = 1$). Models that perform well typically produce data points that cluster closely around this 45-degree reference line.

In the training phase (Fig. 5-A), most models show a strong alignment with the diagonal, indicating high prediction accuracy. Notably, the LM-20-100 and LM-20-125 models demonstrate excellent agreement, with data points tightly concentrated along the ideal line. In contrast, configurations such as LM-15-100, LM-20-75, and LM-10-125 display a greater degree of dispersion, suggesting comparatively lower accuracy during training.

As expected, the testing phase (Fig. 5-B) reveals a broader spread of data points, consistent with the slightly lower R^2 values observed in this phase. Models such as LM-20-75, LM-15-100, and all LM-10 configurations exhibit noticeable deviation from the diagonal line, reflecting reduced generalization performance. Nonetheless, the overall trend remains consistent with the numerical results, further validating the robustness of LM-20-100 and LM-20-125 as the most reliable models across both training and testing datasets.

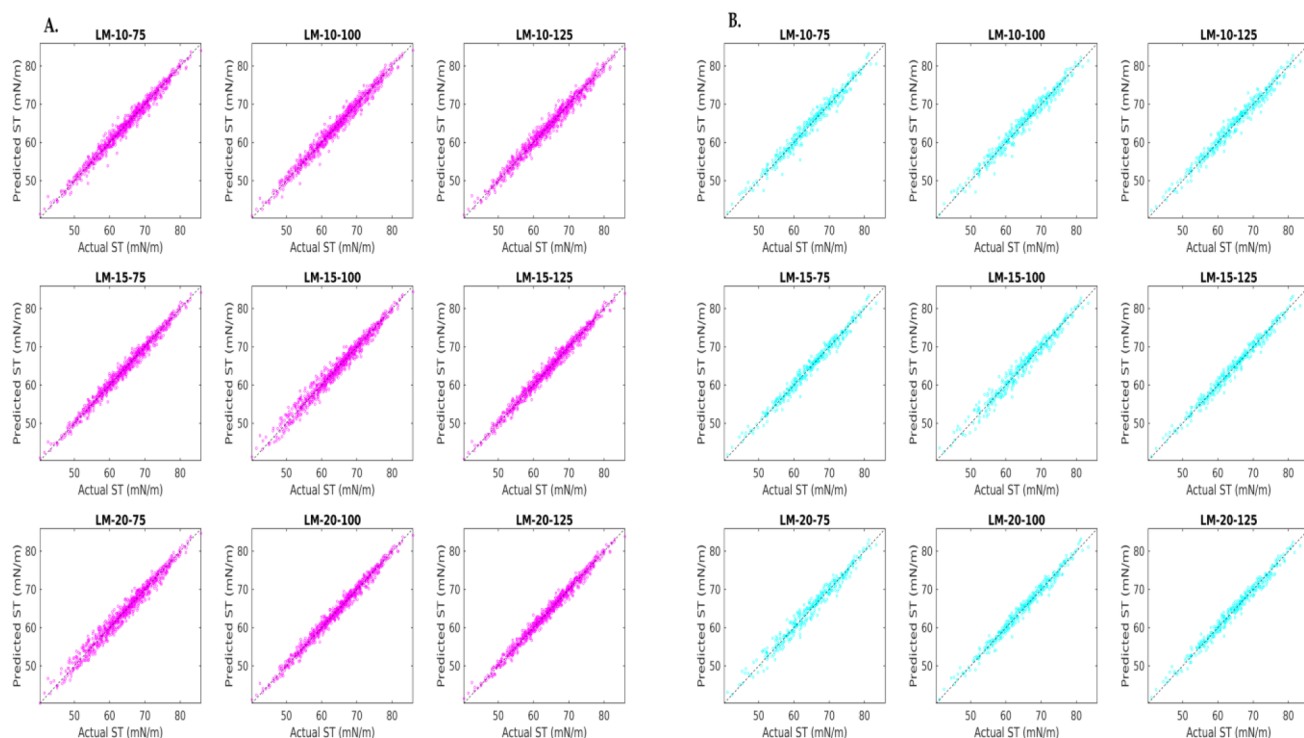


Figure 5—Comparison of actual ST with model predicted ST using LM optimization for training (A) and testing (B).

Fig. 6 illustrates the cross plots for all BR-based model configurations during both training (Fig. 6-A) and testing (Fig. 6-B) phases. Compared to the LM scenarios, the BR models exhibit a greater degree of deviation from the ideal 45-degree reference line, further reinforcing the superior predictive performance of the LM-optimized models.

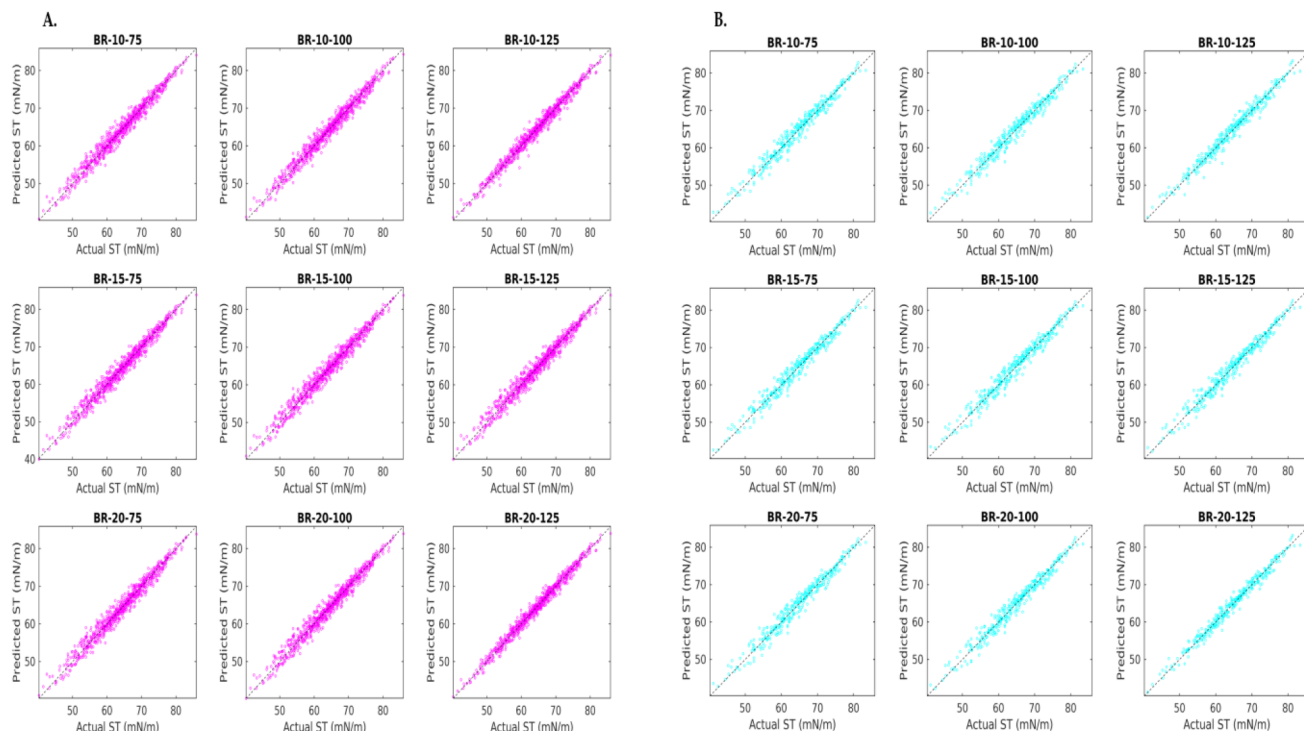


Figure 6—Comparison of actual ST with model predicted ST using BR optimization for training (A) and testing (B).

During the training phase (Fig. 6-A), the BR-20-125 configuration demonstrates the best alignment, with data points closely following the diagonal line, indicating high predictive accuracy. In contrast, the testing phase (Fig. 6-B) reveals a broader spread of data points across all BR configurations. Despite this increased dispersion, the models still maintain reasonable predictive performance.

It is important to note that the visual impact of deviations in the testing phase may appear less pronounced due to the lower number of data points (315) compared to the training phase (735). Nevertheless, the overall comparison highlights the relative effectiveness of LM over BR in minimizing prediction error and improving model consistency across both phases.

Training behavior. The evolution of the loss function, expressed as MSE, is shown across epochs in Fig. 7 for LM-based models and Fig. 8 for BR-based models. These plots reflect the progression of model performance throughout training, with particular emphasis on configurations trained for the same number of epochs but incorporating different numbers of neurons in the hidden layer.

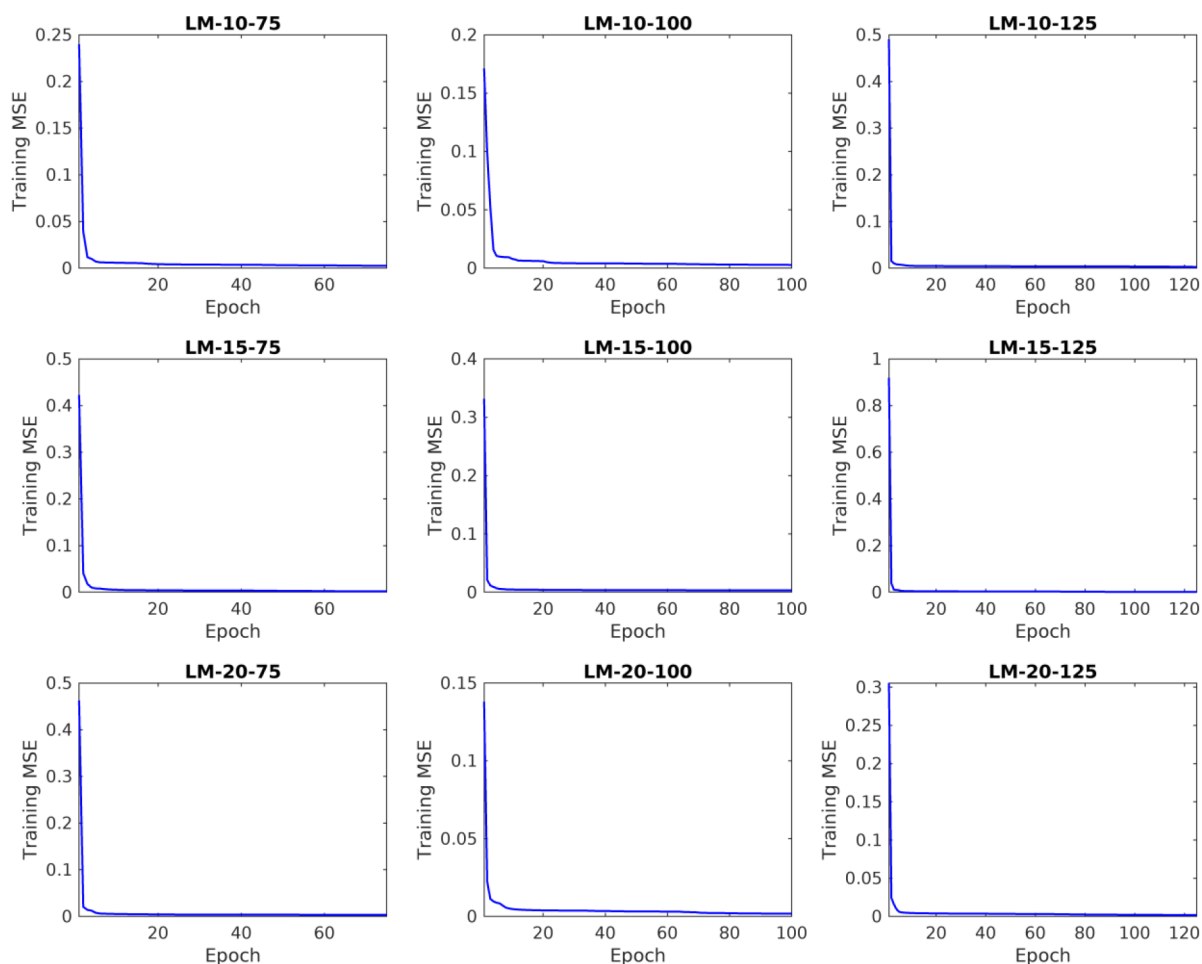


Figure 7—MSE progression over epochs during training with LM optimization.

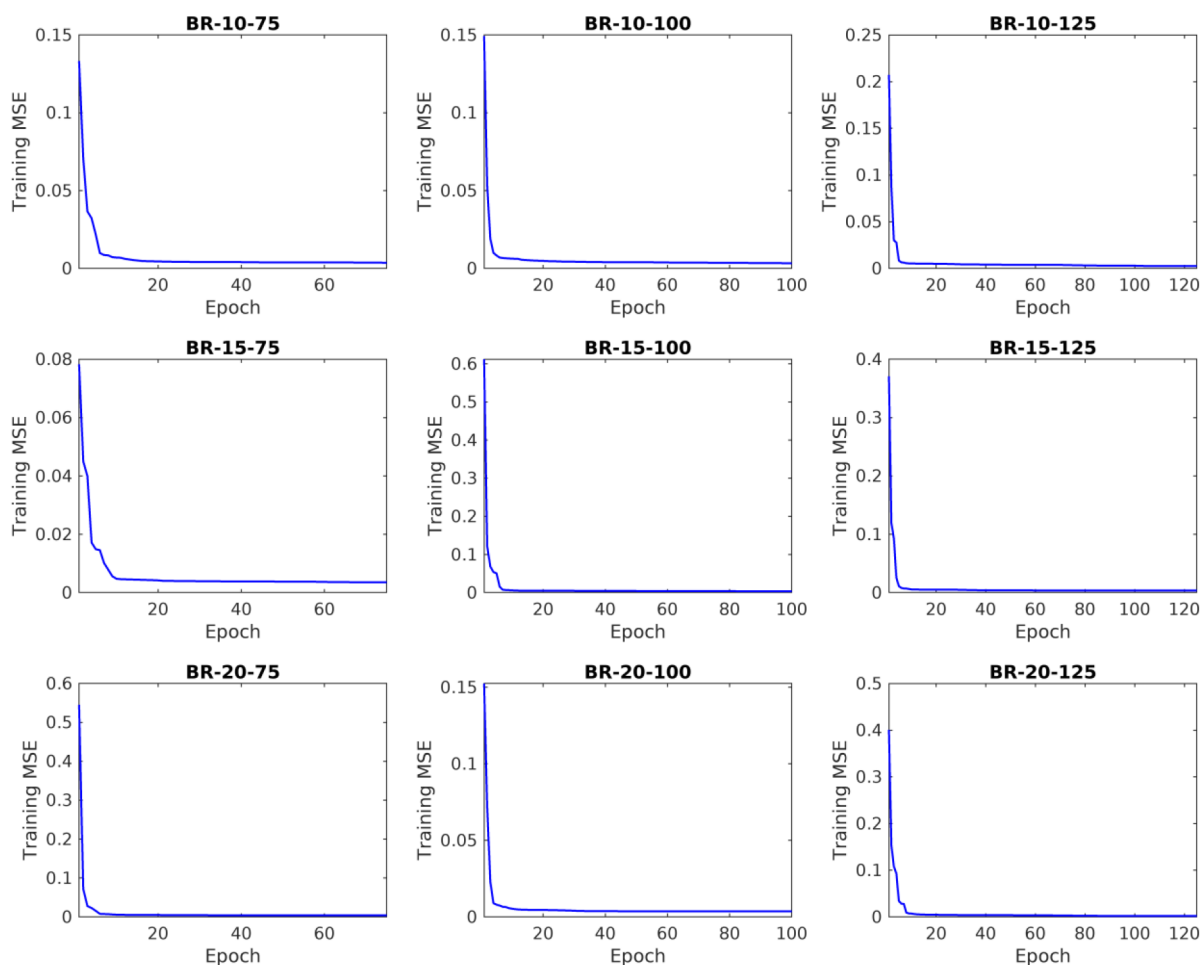


Figure 8—MSE progression over epochs during training with BR optimization.

For the LM models trained over 75 epochs, a sharp decline in MSE is observed during the initial epoch, followed by a plateau as training stabilizes. Among these, the LM-10-75 configuration shows a more gradual error reduction compared to its counterparts with longer training durations (e.g., LM-10-100 and LM-10-125). When the number of epochs is increased to 100, more nuanced training behaviors emerge. For example, LM-10-100 exhibits a three-phase decline: an initial sharp drop during the first few epochs (1–4), a steady reduction around epoch 10, and a further drop near epoch 20. In contrast, LM-15-100 follows a more linear, single-phase decline, similar to the trend observed in the 75-epoch cases. LM-20-100 displays a distinct two-phase profile, with a rapid initial drop and a secondary, less pronounced decline around epoch 6, followed by a subtle adjustment near epoch 78. For LM models trained over 125 epochs, the MSE profiles tend to converge toward consistent behavior, with diminishing gains observed beyond the earlier epochs.

In comparison, the BR-based models in Fig. 8 reveal different training characteristics. The MSE trajectories for BR scenarios often exhibit a stepwise decline pattern, marked by multiple phases of error reduction with varying intensities. For instance, BR-15-75 displays a four-step descent, suggesting intermittent adjustments during training. Similarly, BR-10-75 shows a three-stage decline, with each phase differing in the rate and magnitude of improvement. This behavior reflects the regularization mechanism of the BR algorithm, which balances fitting the training data with minimizing model complexity.

This behavior reflects the distinct characteristics of each optimization technique during the training process. The Levenberg–Marquardt (LM) algorithm typically converges rapidly with sharp and smooth reductions in error, indicating a more aggressive and efficient minimization path. In contrast, Bayesian

Regularization (BR) exhibits a more gradual and multi-phase decline in MSE, which reflects its regularization strategy that prioritizes smoother training and generalization over rapid convergence.

Effect of epochs number. Fig. 9 illustrates the impact of the number of training epochs on model prediction accuracy, as measured by R^2 , for both the training (Fig. 9-A) and testing (Fig. 9-B) phases. In the training phase, a general trend of improved accuracy with increased epoch count is observed across most model configurations. However, certain models—specifically LM-10, LM-15, and BR-15—deviate from this trend. For these configurations, increasing the number of epochs from 75 to 100 resulted in a reduction in R^2 , with the decline being most notable in the LM-15 model. This suggests that beyond a certain point, extended training may lead to diminishing returns or potential overfitting in specific architectures.

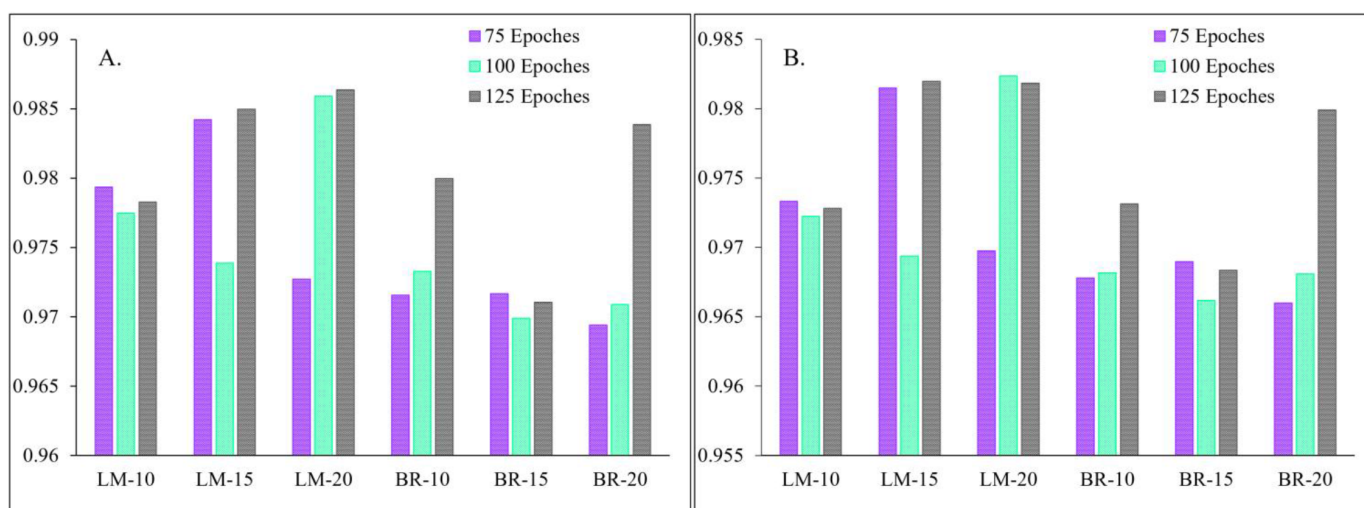


Figure 9—Changes in R^2 of the optimized ANN models as a function of epochs count for training (A) and testing (B).

In contrast, the testing phase (Fig. 9-B) does not exhibit a consistent relationship between the number of epochs and prediction accuracy. The R^2 values fluctuate irregularly with changes in epoch count, indicating that the model's generalization performance is influenced by more than just the number of training iterations. These observations underscore the importance of balanced hyperparameter tuning, where an optimal epoch count must be selected not only for training performance but also to ensure robust testing accuracy.

Effect of neurons number. To evaluate the influence of the number of neurons in the hidden layer on model performance, the R^2 values corresponding to each combination of optimization method and epoch count were plotted against varying neuron counts, as shown in Fig. 10. The training results (Fig. 10-A) indicate that increasing the number of neurons does not universally improve model accuracy. In fact, for certain configurations—such as LM-75, BR-75, and BR-100—the use of 20 neurons resulted in a noticeable decline in R^2 compared to models with only 10 neurons. This suggests that adding complexity to the network can, in some cases, lead to overfitting or hinder convergence during training.

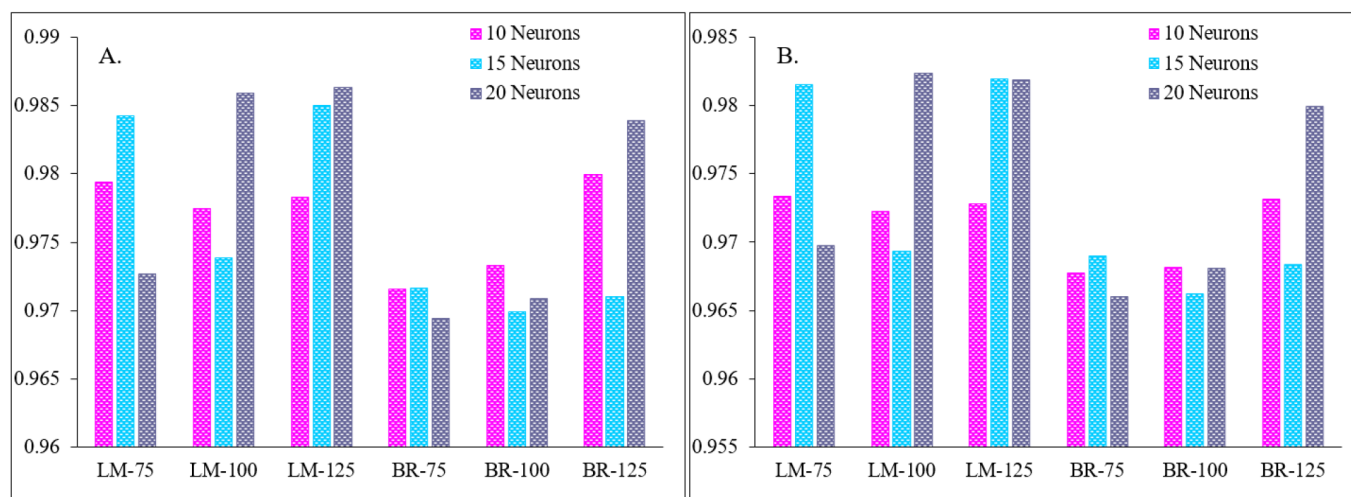


Figure 10—Changes in R^2 of the optimized ANN models as a function of neurons count for training (A) and testing (B).

A similar pattern is observed in the testing phase (Fig. 10-B), where the predictive accuracy does not consistently benefit from a larger neuron count. Notably, the LM-100 configuration exhibits the most significant change in R^2 when increasing the number of neurons from 10 to 20, reflecting a sensitivity to model capacity. These findings emphasize that optimal neuron selection is case-dependent and that larger networks do not necessarily yield better generalization, particularly when not properly balanced with other hyperparameters.

Overall, the sensitivity analysis conducted in this study aimed to optimize the model's predictive performance while providing deeper insights into how variations in key hyperparameters—namely, the number of neurons, number of training epochs, and choice of optimization algorithm—influence model behavior. Among the tested configurations, the Levenberg–Marquardt (LM) algorithm consistently demonstrated superior accuracy in predicting ST values, particularly in both training and testing phases. Nevertheless, the Bayesian Regularization (BR) models also exhibited strong and reliable performance, with all R^2 values exceeding 0.965. These results highlight not only the effectiveness of LM as an optimization technique but also the robustness and generalizability of BR across a range of model configurations.

Conclusions

In this study, machine learning (ML) techniques were applied to model ST between H_2/CH_4 gas mixtures and formation brine under reservoir-relevant conditions. The dataset, obtained from a previous experimental study related to UHS in depleted gas reservoirs, consisted of 1,050 data points encompassing salinity, temperature, pressure, density difference, and corresponding ST values. An initial screening phase was conducted using three artificial neural network (ANN) architectures: feedforward neural network (FFNN), legacy feedforward neural network (LFNN), and pattern recognition neural network (PRNN). Among these, the FFNN demonstrated the best overall performance and was thus selected for detailed sensitivity analysis.

Subsequently, FFNN model was trained using two optimization algorithms: Levenberg–Marquardt (LM) and Bayesian Regularization (BR). Model performance was evaluated using the coefficient of determination (R^2), mean squared error (MSE), and average absolute percentage error (AAPE). All tested configurations achieved strong predictive accuracy, with R^2 values exceeding 0.965 in both training and testing phases.

The highest training R^2 (0.986) was achieved by LM-20-100 and LM-20-125, with LM-20-100 also delivering the best generalization in testing ($R^2 = 0.982$). The lowest training MSE was observed for LM-20-125, while LM-20-100 again performed best during testing. AAPE results further supported LM-20-100 as the most accurate and reliable configuration, with values of 0.0117 and 0.0135 for training

and testing, respectively. Although BR models showed slightly lower performance, they remained robust, with all configurations exceeding 0.965 in testing R^2 .

Overall, LM-20-100 was identified as the best-performing model for predicting ST in H_2/CH_4 -brine systems, offering an optimal balance between accuracy and computational efficiency. The strong performance of both LM and BR models highlights the effectiveness of ML-based approaches in modeling interfacial properties relevant to UHS in depleted gas reservoirs.

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